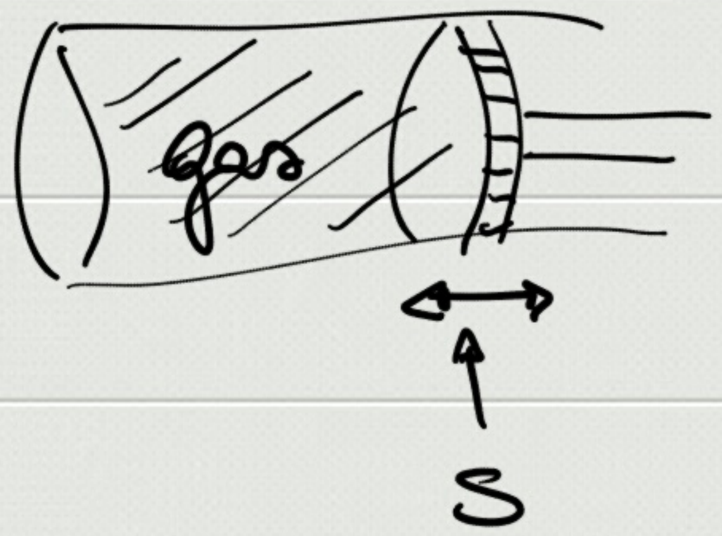
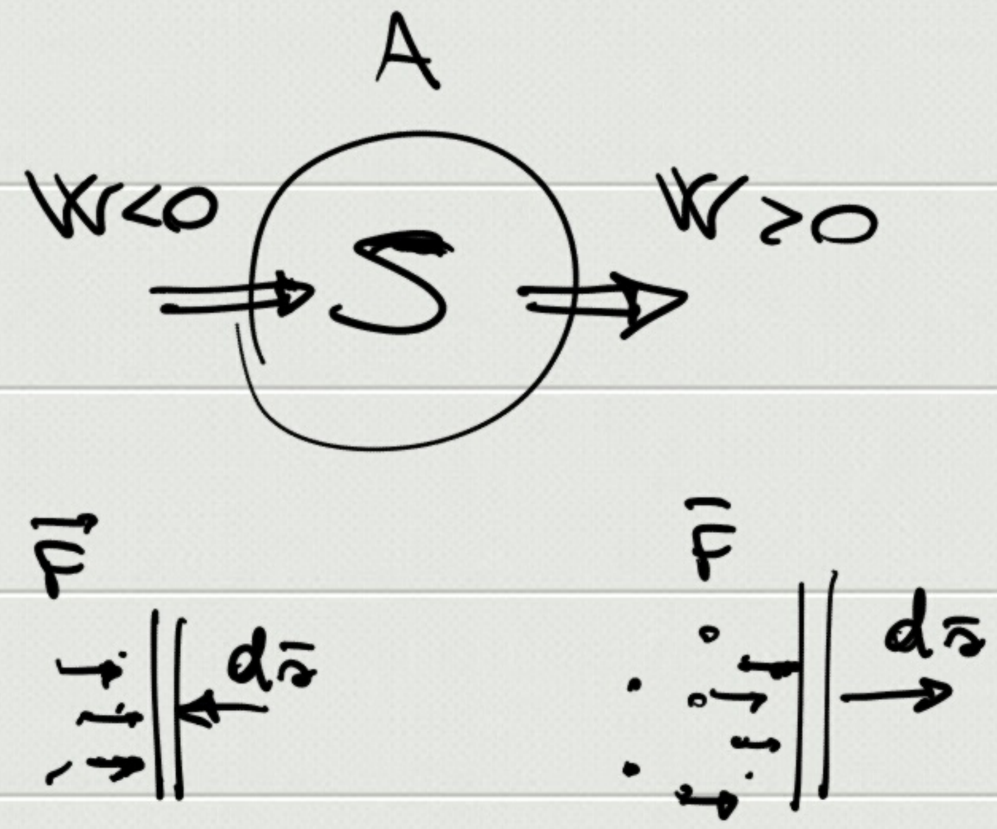
$$dW = \bar{F}\, d\bar{s}$$

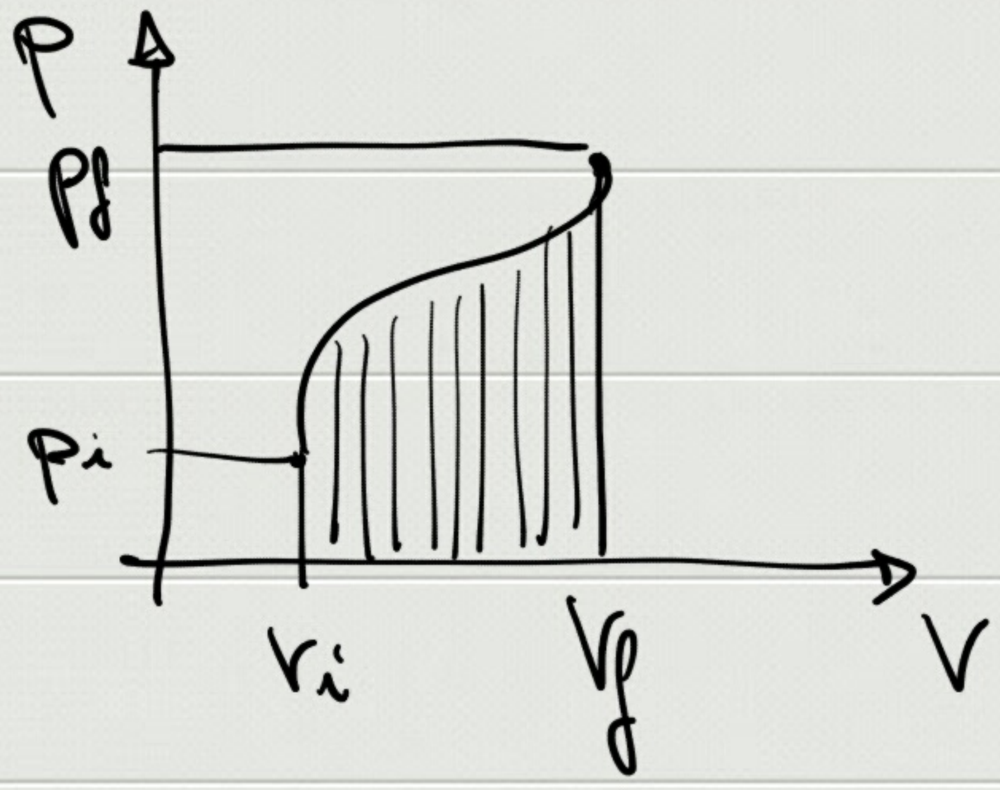




$$\delta W = \bar{F}\, d\bar{s} = p S\, ds = p\, dV$$

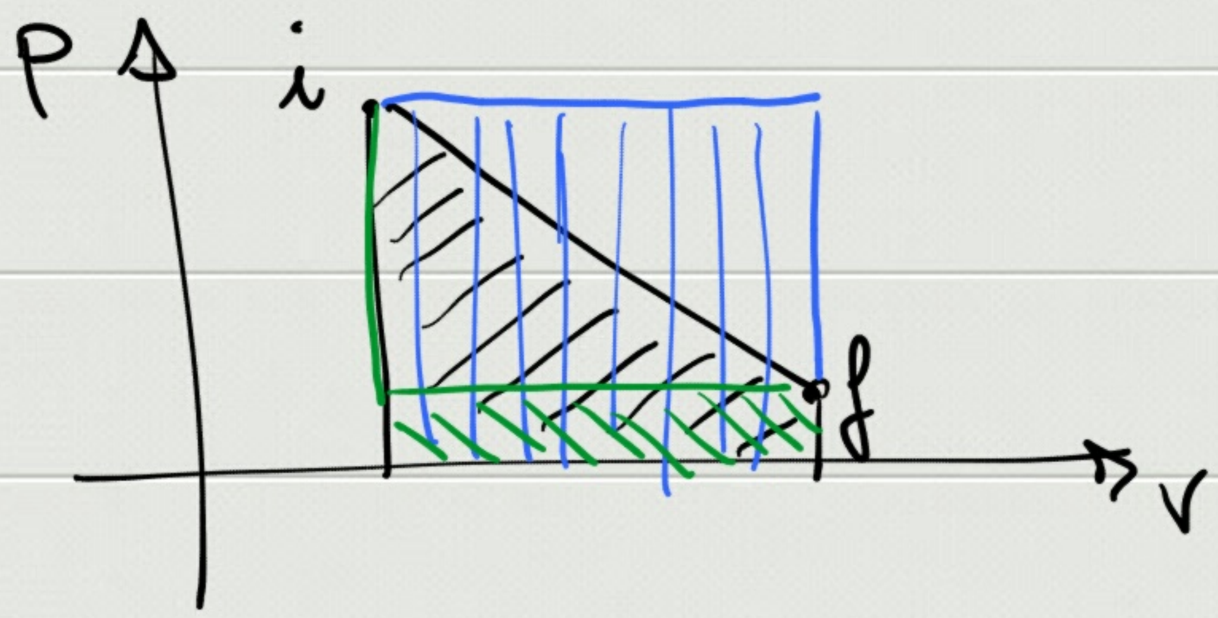
$$W_{i\to f} = \int_i^f p\, dV = \begin{cases} \text{caotico} \Rightarrow & ? \\ \text{quasi statico} \Rightarrow & \underbrace{\int_{V_i}^{V_f} p(V,T)\, dV} \end{cases}$$

$$f(p,V,T) = 0 \Rightarrow p = p(V,T)$$



Clapeyron

$$W_{i\to f} = \int_{v_i}^{v_f} p \, dV$$



$$W_{i\to f} = \int_{v_i}^{v_f} p(v,T)\, dV \neq f(v_f) - f(v_i)$$

differenziale del lavoro non è "esatto"

$$\Rightarrow \boxed{\delta W}$$

$$\Rightarrow \boxed{\delta W = p\, dV}$$